The MABEL Base Motion Model: Swerve Kinematics, Command Smoothing, and Direction-Aware Tip-Over Safety

MABEL Simulation Project*

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Abstract

We derive and experimentally verify the mobile-base motion model of MABEL, a bimanual robot on a three-module swerve base carrying a vertical lift. We cover (i) the swerve inverse kinematics that turn a body-frame twist into per-module steering and wheel commands; (ii) trapezoidal command smoothing; and (iii) a direction-aware tip-over safety layer. Braking creates an inertial moment that can flip the robot; we bound it with the zero-moment-point / support-polygon criterion, making the safe speed and acceleration shrink with lift height and become direction-dependent (forward throttled, lateral/rear agile). For combined motion we constrain the full center-of-mass acceleration—including the Coriolis and centripetal terms about the instantaneous center of rotation—and project the velocity target into the tip-safe envelope. Experiments quantify the safe envelope versus lift, the combined-motion tip map, and twist-tracking fidelity.

Keywords: Mobile robots, swerve drive, tip-over stability, zero-moment point, motion control.

1 Overview

This report derives how a body-frame velocity command becomes wheel commands (swerve inverse kinematics), how commands are smoothed, and how the speed / acceleration limits are tied to the lift height and travel direction so the robot cannot tip over while stopping. Lateral and rearward motion are kept *agile*; only forward braking is throttled. The components map to the implementation as: swerve IK (`mabel/swerve.py`); velocity smoothing + tip caps (`config.safe_base_step`, `teleop_keyboard.py`); physical constants and the safety model (`mabel/config.py`); actuator/hardware specs (`models/mabel_full.xml`).

2 Frames and notation

We use MABEL's chassis (body) frame: origin at the chassis center, x along the chassis $+X$ axis, z up. Mechanically, MABEL's **front is** $-X$ and its **left is** $-Y$.

*Companion notes: whole-body IK (`whole_body_control.pdf`), arm control (`arm_control.pdf`), compliance study (`compliance_study.pdf`).

Symbol	Meaning	Units
$\mathbf{v} = (v_x, v_y)$	commanded body-frame linear velocity	m/s
ω	commanded body-frame yaw rate	rad/s
$\mathbf{r}_i = (r_{ix}, r_{iy})$	module i position in chassis frame	m
θ_i	steering (azimuth) joint angle of module i	rad
ϕ_i	mount yaw offset of module i	rad
R_w	wheel radius	m
g	gravity	m/s^2
h	center-of-mass (CoM) height	m
d_{dir}	horizontal CoM $\rightarrow$ tipping-edge lever	m
z	lift command (travel)	m

The three swerve modules sit at

$$\mathbf{r}_{\text{fl}} = (-0.17, -0.17), \quad \mathbf{r}_{\text{fr}} = (-0.17, +0.17), \quad \mathbf{r}_{\text{b}} = (+0.15,\, 0)\ [\text{m}],$$

i.e. two front wheels ($-X$) and a single rear wheel ($+X$): a triangular "delta" base.

3 Swerve inverse kinematics

3.1 Per-module velocity

The base is a rigid body, so module i moves with the body velocity plus the rotational term $\boldsymbol{\omega} \times \mathbf{r}_i$ with $\boldsymbol{\omega} = \omega \hat{z}$:

$$\mathbf{v}_i = \begin{bmatrix} v_{ix} \\ v_{iy} \end{bmatrix} = \begin{bmatrix} v_x - \omega\, r_{iy} \\ v_y + \omega\, r_{ix} \end{bmatrix}. \tag{1}$$

3.2 Wheel speed and heading

Each module points its wheel along $\mathbf{v}_i$ and spins to match its speed:

$$\psi_i = \operatorname{atan2}(v_{iy}, v_{ix}), \qquad \dot{\varphi}_i = \frac{\|\mathbf{v}_i\|}{R_w}, \qquad \theta_i^{\text{target}} = \psi_i - \phi_i. \tag{2}$$

Origin of ϕ_i. In the model the wheel body is mounted with `xyaxes="0 0 1 0 1 0"`, which puts the drive axle along the steer frame's X and makes the wheel roll along the steer frame's $+Y$ — i.e. $+\frac{\pi}{2}$ from the steer-frame X. Adding the housing mount yaw (the XML `euler` Z) gives

$$\boxed{\phi_i = \text{euler}_i + \tfrac{\pi}{2}} \tag{3}$$

Sanity check: for a pure spin ($\mathbf{v} = 0$, $\omega \neq 0$) all three θ_i^{target} collapse to the same value, because every module's mount frame is oriented toward the chassis center.

3.3 Shortest-path steering (module flip)

A steering wheel is **π-periodic**: pointing at θ and driving forward equals pointing at $\theta + \pi$ and driving in reverse. We pick the representative closest to the current angle, choosing the drive sign $s \in \{+1, -1\}$. First wrap the error to $(-\pi, \pi]$,

$$\Delta = \big((\theta^{\text{target}} - \theta^{\text{cur}} + \pi) \bmod 2\pi\big) - \pi, \tag{4}$$

then take the shorter of steering vs. reversing:

$$(\theta^\star, s) = \begin{cases} (\theta^{\text{cur}} + \Delta - \pi,\ -1), & \Delta > \frac{\pi}{2}, \\ (\theta^{\text{cur}} + \Delta + \pi,\ -1), & \Delta < -\frac{\pi}{2}, \\ (\theta^{\text{cur}} + \Delta,\ +1), & \text{otherwise.} \end{cases} \tag{5}$$

The column never sweeps more than $\pm\frac{\pi}{2}$.

Continuous steer joint. Because the steer joint is continuous (`limited="false"`), its position actuator must be unlimited too (`ctrllimited="false"`). A $\pm 2\pi$ clamp would let the accumulated shortest-path angle saturate, sticking a module at the wrong heading — a real bug, fixed in `mabel_full.xml`.

3.4 Anti-slip drive choke

Driving before the wheel has turned scrubs the floor, so we scale wheel speed by cos of the steering error and cut power beyond a deadband θ_{db} ($0.52\,\text{rad} \approx 30°$):

$$\dot{\varphi}_i^{\text{cmd}} = \frac{\|\mathbf{v}_i\|}{R_w}\, s \cdot \begin{cases} \cos\big(|\theta^\star - \theta^{\text{cur}}|\big), & |\theta^\star - \theta^{\text{cur}}| \le \theta_{\text{db}}, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

4 Hardware: 3 in REV MAXSwerve (REV-21-3005)

The actuator limits in `mabel_full.xml` are matched to the real module.

	Drive	**Steer (azimuth)**
Motor	NEO	NEO 550
Free speed	5676 rpm	11 000 rpm
Stall torque	2.6 N m	0.97 N m
Gear ratio	4.71:1 (high)	46.42:1 (9424:203)
Wheel	3 in dia. $\Rightarrow R_w = 0.0381$ m	
Module limit	$\sim$126 rad/s ($\approx$4.8 m/s)	$\sim$25 rad/s azimuth
MuJoCo `forcerange`	$\pm$11 N m	$\pm$32 N m

Roles. The **steering** actuators are tuned for speed (low k_v, full torque) so modules reorient with minimal latency — this is what makes left/right strafing feel agile. The **drive** actuators are where safety lives: their acceleration and speed are governed by the tip-over model below, not by the raw motor limit.

5 Command smoothing (acceleration ramp)

Raw teleop targets $(\tilde{v}_x, \tilde{v}_y, \tilde{\omega})$ are rate-limited so the velocity profile is continuous (trapezoidal). For a linear axis with cap $a_{\max}$ and timestep $\mathrm{d}t$:

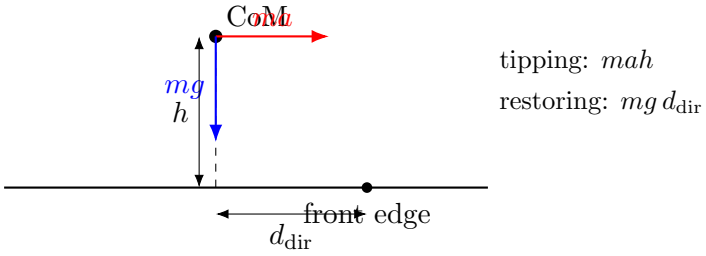
$$v_x \leftarrow \begin{cases} \min(\tilde{v}_x,\ v_x + a_{\max}\mathrm{d}t), & v_x < \tilde{v}_x, \\ \max(\tilde{v}_x,\ v_x - a_{\max}\mathrm{d}t), & v_x \ge \tilde{v}_x. \end{cases} \tag{7}$$

Crucially $a_{\max}$ is *not* constant — it is the tip-safe limit for the current direction and lift height (next section).

6 Direction-aware tip-over model

6.1 Tipping condition

While the base brakes at magnitude a, an inertial (d'Alembert) force ma acts at the CoM in the travel direction. About the support edge it is pushing toward, this makes a tipping moment mah; gravity resists with $mg\,d_{\text{dir}}$, where d_{dir} is the horizontal CoM-to-edge lever in that direction.



Stability requires the restoring moment to dominate:

$$\boxed{m\,a\,h \le m\,g\,d_{\text{dir}}} \quad \Longrightarrow \quad a \le a_{\text{tip}}(h) = \frac{g\,d_{\text{dir}}}{h}. \tag{8}$$

The mass m **cancels**: tipping depends only on geometry and g.

6.2 Per-direction levers

Measured from `mabel_sim.xml` (CoM via `subtree_com`, support polygon from wheel-contact points; CoM at $x \approx -0.06$, slightly front-heavy):

Direction	Lever d_{dir} [m]	Tipping edge
Forward ($-X$)	0.094	front edge (fl–fr)
Lateral ($\pm Y$)	0.103	side edges (fl–b / fr–b)
Rearward ($+X$)	0.225	rear wheel (b)

The triangular base with a single rear wheel makes **forward** the binding direction; lateral is only marginally larger, rearward is generous. Hence agile sideways motion comes mainly from *fast steering*, while the forward speed is the one genuinely throttled.

6.3 CoM height vs. lift

A fixed low base plus an upper body that rises with the lift command z gives a CoM height affine in z:

$$h(z) = \frac{m_b h_b + m_u(h_{u0} + z)}{m_b + m_u} = h_0 + k\,z. \tag{9}$$

Fit to the model (conservative — never under-estimates h):

$$h(z) \approx 0.443 + 0.432\,z \text{ [m]}, \qquad z \in [0, 0.635]\,\text{m}, \tag{10}$$

so h ranges from 0.443 m (down) to 0.717 m (fully up).

6.4 Safe acceleration and speed (pure-axis)

Each direction gets its own safety factor: forward tipping is the genuine hazard and stays conservative, while lateral/rearward tips have more margin *and* are far less dangerous, so they run near the tip limit for snappy braking:

$$\boxed{a_{\text{safe}}^{\text{dir}}(z) = \frac{g\, d_{\text{dir}}}{\text{SF}_{\text{dir}}\left(h_0 + kz\right)}}, \tag{11}$$

with $\text{SF}_{\text{fwd}} = 2.0$ and $\text{SF}_{\text{lat}} = \text{SF}_{\text{back}} = 1.2$. This a_{safe} is exactly the ramp cap $a_{\max}$, so a hard stop never exceeds the tip-safe rate. With a per-direction stopping time Δt_{dir}, a stop from v needs $a = v/\Delta t$, giving the speed cap

$$\boxed{v_{\max}^{\text{dir}}(z) = a_{\text{safe}}^{\text{dir}}(z)\,\Delta t_{\text{dir}} = \frac{g\, d_{\text{dir}}\,\Delta t_{\text{dir}}}{\text{SF}_{\text{dir}}\,(h_0 + kz)}}, \tag{12}$$

so that $v_{\max} \propto d_{\text{dir}}/h$. **Max speed scales linearly with the support lever and inversely with the CoM height.** The *full-speed reversal time* is $\approx 2\,\Delta t_{\text{dir}}$, so Δt is the agility knob. Forward keeps a long, gentle stop ($\Delta t_{\text{fwd}} = 0.8\,\text{s}$) and a conservative ceiling (0.8 m/s); lateral/rear use a short stop ($\Delta t_{\text{agile}} = 0.55\,\text{s}$) and an agile ceiling (1.5 m/s). Caps are clamped to $[V_{\text{floor}}, V_{\text{ceil}}]$.

6.5 Combined motion: ICR, CoM acceleration, and the ZMP criterion

The pure-axis caps of §6.4 assume *one* acceleration component at a time. Under a combined twist (v_x, v_y, ω) this is unsafe. With $\omega \neq 0$ the base rotates about its **instantaneous centre of rotation** (ICR), the body point of zero velocity,

$$\mathbf{r}_{\text{ICR}} = \big(-v_y/\omega,\ v_x/\omega\big), \tag{13}$$

and every body point — including the CoM at offset $\mathbf{r}_c$ — carries a centripetal acceleration toward the ICR. By the transport theorem the CoM horizontal acceleration in the body frame is

$$\boxed{\mathbf{a}_c = \underbrace{\begin{bmatrix}\dot v_x - \omega v_y \\ \dot v_y + \omega v_x\end{bmatrix}}_{\text{transport / Coriolis}} + \underbrace{\dot\omega\,\hat z \times \mathbf{r}_c}_{\text{tangential}} - \underbrace{\omega^2\,\mathbf{r}_c}_{\text{centripetal}}}. \tag{14}$$

The Coriolis term $\omega\,(v_x, v_y)$ is the cause of "it tips when I turn while driving": e.g. $\omega = 2\,\text{rad/s}$ at $v = 0.8\,\text{m/s}$ injects $1.6\,\text{m/s}^2$ of *lateral* CoM acceleration that no per-axis cap sees.

Zero-moment-point criterion. For a CoM at height h on flat ground, the gravity+inertial resultant pierces the ground at the zero-moment point

$$\mathbf{x}_{\text{zmp}} = \mathbf{x}_c - \frac{h}{g}\,\mathbf{a}_c \tag{15}$$

(the cart-table relation; it neglects the CoM vertical acceleration and the body's angular-momentum rate). The base keeps full ground contact *iff* $\mathbf{x}_{\text{zmp}}$ lies inside the support polygon $\mathcal{P}$ — the triangle of the three wheel contacts. **The edge $\mathbf{x}_{\text{zmp}}$ first crosses is the direction the robot falls.**

Acceleration polytope. Substituting (15), the criterion is a *linear* constraint on $\mathbf{a}_c$: for each edge k with outward unit normal $\hat{o}_k$ and CoM lever d_k,

$$\boxed{(-\hat{o}_k)\cdot\mathbf{a}_c \;\le\; \frac{g\,d_k}{\mathrm{SF}_k\,h} \qquad \forall k.} \tag{16}$$

This is exactly the single-edge bound of §6.1, but applied to the *total* $\mathbf{a}_c$ from (14), so it couples v_x, v_y, ω and their rates; the pure-axis caps of §6.4 are the special case where only one component of (14) is nonzero. Equation (16) is the discrete, flat-ground form of the *force–angle stability measure* (Papadopoulos & Rey, 1996) and the *ZMP* margin (Vukobratović); the stability margin is $\min_k \left[\, gd_k/(\mathrm{SF}_k h) - (-\hat{o}_k)\cdot\mathbf{a}_c \,\right]$.

6.6 Control law (body frame): envelope projection

We enforce (16) by projecting the *velocity target* into the tip-safe envelope before the per-axis ramp, rather than gating the command change. At zero command acceleration the CoM acceleration $\mathbf{a}_c = (-\omega v_y - \omega^2 r_x,\ \omega v_x - \omega^2 r_y)$ is **degree-2 homogeneous** in the twist (v_x, v_y, ω), and the usage $u = \max_k(-\hat{o}_k\cdot\mathbf{a}_c)/(gd_k/\mathrm{SF}_k h)$ is linear in $\mathbf{a}_c$, so $u(s\cdot\text{twist}) = s^2\,u(\text{twist})$. The largest scale keeping the steady ZMP inside $\mathcal{P}$ is therefore closed-form,

$$\boxed{s = \min\!\left(1,\ u^{-1/2}\right), \qquad (\tilde{v}_x, \tilde{v}_y, \tilde{\omega}) \leftarrow s\,(\tilde{v}_x, \tilde{v}_y, \tilde{\omega}),} \tag{17}$$

applied to the operator target $(\tilde{v}_x, \tilde{v}_y, \tilde{\omega})$; the result is then tracked by the per-axis trapezoidal ramp of §6.4 (forward braking throttled, lateral/rear agile), whose bounded rate also limits the brake when an over-limit twist is pulled back in. This caps turn-while-drive (and pure over-spin) at the support-polygon boundary while leaving pure-axis motion at the agile per-axis limits. Implemented as `com_accel`, `tip_usage`, `tip_speed_scale` in `mabel/config.py`, applied in `teleop_keyboard.py`.

Critical note (why projection, not a blend). An earlier version blended the previous twist toward the candidate by the largest $\beta \in [0,1]$ keeping the ZMP feasible. That **dead-locks**: a fast spin mixed with linear velocity is a state whose *current* twist already violates (16) (the spin alone tips forward above $\omega \approx \sqrt{gd_{\mathrm{fwd}}/(\mathrm{SF}h\,|r_{c,x}|)} \approx 4.2\,\mathrm{rad/s}$, since the CoM leads the spin centre), so *every* β is infeasible and the fallback $\beta = 0$ means "hold velocity" — the robot keeps spinning and cannot stop, because braking is itself an acceleration the constraint forbids. Projecting the *target* (17) avoids this: the stop target 0 has $u = 0 \Rightarrow s = 1$, so a stop is never throttled, and an over-limit state is ramped back in rather than frozen.

6.7 Dual view — "same deceleration"

If instead the deceleration is held at a fixed a_d, the constraint $a_d \le gd_{\mathrm{dir}}/(\mathrm{SF}_{\mathrm{dir}}\,h(z))$ becomes a limit on how high you may raise the lift,

$$z^\star = \frac{1}{k}\left(\frac{g\,d_{\mathrm{dir}}}{\mathrm{SF}_{\mathrm{dir}}\,a_d} - h_0\right), \tag{18}$$

above which a_d must shrink (a smoother, longer stop). With a fixed a_d and a stopping-distance budget $L = v^2/(2a_d)$, the speed limit is $v_{\max} = \sqrt{2a_d L} \propto 1/\sqrt{h}$. Same physics; the controller uses the fixed-Δt form above.

7 Worked numbers

$g = 9.81\,\mathrm{m/s^2}$, $\mathrm{SF}_{\mathrm{fwd}} = 2.0$, $\mathrm{SF}_{\mathrm{agile}} = 1.2$, $\Delta t_{\mathrm{fwd}} = 0.8\,\mathrm{s}$, $\Delta t_{\mathrm{agile}} = 0.55\,\mathrm{s}$, $h(z) = 0.443 + 0.432z$, levers $(d_{\mathrm{fwd}}, d_{\mathrm{lat}}, d_{\mathrm{back}}) = (0.094, 0.103, 0.225)\,\mathrm{m}$.

Lift z [m]	$h(z)$ [m]	v_{fwd}	v_{lat} [m/s]	v_{back}	a^{fwd}	a^{lat} [m/s^2]	a^{back}
0.000	0.443	0.80	1.05	1.50	1.04	1.90	4.15
0.200	0.529	0.70	0.88	1.50	0.87	1.59	3.48
0.400	0.616	0.60	0.75	1.50	0.75	1.37	2.99
0.635	0.717	0.51	0.65	1.41	0.64	1.17	2.56

Forward stays conservative and shrinks $\sim$40% as the lift rises; lateral and rearward decelerate $\sim 1.7\times$ harder for agile reversals. The full-speed lateral reversal takes $\approx 2\Delta t_{\mathrm{agile}} = 1.1\,\mathrm{s}$ (vs. 1.6 s under the old uniform $\mathrm{SF} = 2$, $\Delta t = 0.8\,\mathrm{s}$).

Combined motion. The coupled criterion (16) shows why turn-while-drive is the real hazard. A steady forward + yaw twist (no commanded ramp, $\mathbf{a}_c = \omega(v_x, v_y)^{\perp} - \omega^2 \mathbf{r}_c$) gives the support-polygon usage $u = \max_k(-\hat{o}_k \cdot \mathbf{a}_c)/(g d_k/\mathrm{SF}_k h)$:

(v, ω)	$0.8\,\mathrm{m/s}, 2\,\mathrm{rad/s}$	$1.5, 2$	$1.5, 3$	$0.8, 2$
lift z	0	0	0	0.635
u	0.65	1.28	1.87	1.05
	safe	throttled	throttled	throttled

At the lowered lift, $0.8\,\mathrm{m/s} + 2\,\mathrm{rad/s}$ is borderline (u=0.65); raising the lift to 0.635 m pushes the *same* twist to u=1.05, so the envelope projection (17) scales it back (s=$u^{-1/2}$) — matching the observed tilt when combining motions with the body raised.

> For reference, the previous constant `MAX_LIN_ACCEL`= $2.0\,\mathrm{m/s^2}$ exceeded a_{tip} for nearly the entire lift range, and the per-axis caps ignored $\mathbf{a}_c$'s ωv coupling entirely — a hard forward stop with the lift raised, or a fast turn while driving, could flip the robot. The direction-aware caps and the ZMP polytope (16) fix both.

8 Parameters (`mabel/config.py`)

Symbol	Constant	Default
$d_{\mathrm{fwd}}, d_{\mathrm{lat}}, d_{\mathrm{back}}$	`TIP_LEVER_{FWD,LAT,BACK}`	$0.094, 0.103, 0.225\,\mathrm{m}$
$\mathrm{SF}_{\mathrm{fwd}}, \mathrm{SF}_{\mathrm{agile}}$	`TIP_SF_FWD`, `TIP_SF_AGILE`	2.0, 1.2
$\Delta t_{\mathrm{fwd}}, \Delta t_{\mathrm{agile}}$	`STOP_TIME_FWD`, `STOP_TIME_AGILE`	0.80 s, 0.55 s
h_0, k	`COM_Z0`, `COM_DZDLIFT`	0.443, 0.432
V_{floor}	`V_SPEED_FLOOR`	$0.15\,\mathrm{m/s}$
$V_{\mathrm{ceil}}^{\mathrm{fwd}}$	`V_FWD_CEIL`	$0.80\,\mathrm{m/s}$
$V_{\mathrm{ceil}}^{\mathrm{agile}}$	`V_AGILE_CEIL`	$1.50\,\mathrm{m/s}$
g	`GRAVITY`	9.81

Helpers: `com_height(z)`, `max_decel_dir(z)`, `max_speed_dir(z)` (pure-axis), and `com_accel`, `tip_usage`, `tip_speed_scale`, `accel_safe_scale`, `safe_base_step` (coupled), plus the support polygon `SUPPORT_PTS` / `COM_XY`.

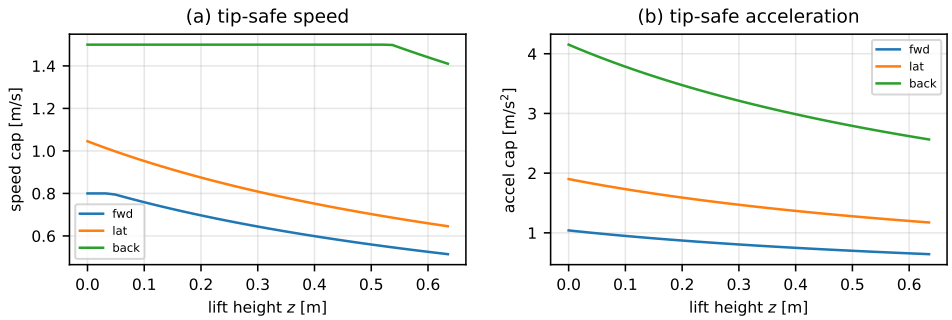


Figure 1: Tip-safe (a) speed and (b) acceleration caps vs. lift height, per direction. Caps fall as the center of mass rises; forward is throttled most.

9 Experiments

All experiments run in MuJoCo on the full MABEL model (`tools/doc_experiments.py`, `tools/compliance_study.py`).

9.1 Safe envelope vs. lift height

Fig. 1 plots the direction-aware speed and acceleration caps as the lift rises. Both shrink roughly as $1/h(z)$: forward is the tightest (front lever), lateral slightly larger, rearward the most generous, confirming the analysis of §6.4.

9.2 Combined-motion tip map

Fig. 2 maps the peak chassis tilt over simultaneous linear speed and yaw rate at full lift—the binding case. The 5° contour delimits the safe operating region; beyond it the quasi-static cap is optimistic because the cart-table ZMP omits the spinning body's angular-momentum rate (§9 motivates the conservative envelope projection used online).

9.3 Twist tracking

Fig. 3 commands a forward+yaw twist step and back to zero; the swerve base realizes the commanded body velocity and yaw rate with the trapezoidal, tip-limited profile and stops cleanly, verifying (1) and the smoother.

10 Assumptions & limitations

- **Quasi-static rigid tip-over.** Suspension compliance, wheel slip, and payload sloshing are ignored; SF_k absorbs them.
- **Cart-table ZMP.** (15) neglects the CoM *vertical* acceleration and the body's angular-momentum rate — notably the reaction of swinging arms / a moving lift, which a full ZMP would include via $\dot{\mathbf{L}}$. $\mathbf{r}_c, h$ are taken at the current lift but assumed otherwise fixed.
- **Flat ground.** (15)–(16) assume a level support plane; on a slope the gravity term tilts and $\mathcal{P}$ must be projected accordingly.

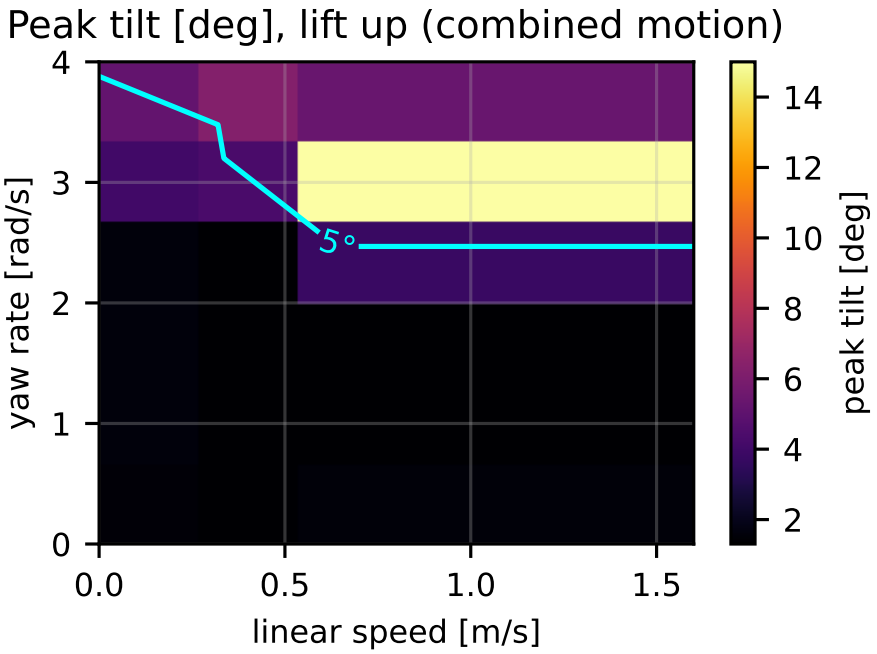


Figure 2: Peak chassis tilt under combined linear+yaw motion at full lift; the 5° contour bounds the safe envelope.

- **Trapezoidal profile.** Acceleration is piecewise-constant (jerk unbounded at corners); an S-curve would further smooth onset.
- **First-order base model.** The smoother shapes the *commanded* twist; it assumes the swerve servos realize it within a tick (no wheel-slip or traction limit modelled beyond the anti-slip choke of §2.4).

Related work

The criterion (16) is the flat-ground, point-contact specialization of two standard mobile-robot tip-over measures: the *zero-moment point* and its support-polygon test (Vukobratović & Borovac, 2004), and the *force–angle stability measure* (Papadopoulos & Rey, *ICRA* 1996), which scores stability by the angle between the net force at the CoM and the lines to the support-polygon edges. Related dynamic metrics include the moment-height stability measure (Moosavian & Alipour) and the dynamic stability margin. We use the linear acceleration-polytope form because it is cheap to evaluate online and composes directly with the velocity smoother.

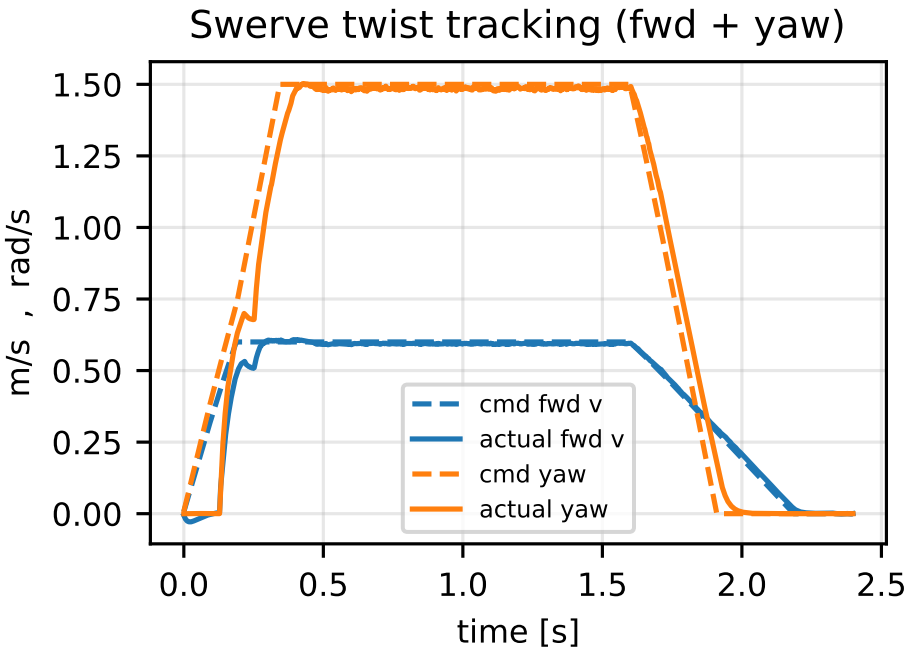


Figure 3: Commanded vs. achieved forward velocity and yaw rate for a combined twist step; the base tracks and stops under the tip-safe smoother.